

Article

Deep Learning in Financial Modeling: Predicting European Put Option Prices with Neural Networks

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Abstract: This paper explores the application of deep neural networks (DNNs) as an alternative to the traditional Black–Scholes model for predicting European put option prices. Using synthetic datasets generated under the Black–Scholes framework, the proposed DNN achieved strong predictive performance, with a Mean Squared Error (MSE) of 0.0021 and a coefficient of determination (R^2) of 0.9533. This study highlights the scalability and adaptability of DNNs to complex financial systems, offering potential applications in real-time risk management and the pricing of exotic derivatives. While synthetic datasets provide a controlled environment, this study acknowledges the challenges of extending the model to real-world financial data, paving the way for future research to address these limitations.

Keywords: deep learning; artificial intelligence; neural networks; European options; option pricing; Black–Scholes model; financial prediction; quantitative finance; computational finance



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1. Introduction

Option pricing is a cornerstone of quantitative finance, providing tools for risk management and derivative valuation. European put options, in particular, play a critical role in hedging strategies and investment decision-making. Among the most widely used models, the Black–Scholes framework, introduced in 1973 [1], provides an elegant and tractable closed-form solution. However, its reliance on assumptions such as constant volatility, frictionless markets, and normal asset return distributions limits its effectiveness in capturing the nonlinear and stochastic dynamics of modern financial markets [2].

In recent years, advances in deep learning have shown promise in addressing these challenges. Neural networks, with their ability to model nonlinear relationships and capture complex patterns in large datasets, present a flexible alternative to traditional methods. Unlike the B&S model, neural networks do not require restrictive assumptions about market behavior, making them well-suited for scenarios involving noisy data and dynamic environments. Several studies have applied deep learning to financial modeling, including physics-informed neural networks (PINNs) and Fourier Neural Operators (FNOs), which integrate domain knowledge into neural network architectures to solve partial differential equations (PDEs) and stochastic differential equations (SDEs) [3–6]. While these methods have demonstrated strong theoretical potential, they often require significant computational resources and can struggle with convergence in high-dimensional problems.

This study proposes a dense neural network (DNN) framework for pricing European put options. Compared to PINNs and FNOs, our approach emphasizes computational simplicity and scalability, making it suitable for large-scale applications. The key objective

is to leverage synthetic datasets generated using the B&S analytical solution to train and validate the model, providing a controlled environment for evaluating its performance. The DNN's ability to approximate theoretical prices with high accuracy underpins its potential for extension to real-world financial data. While the reliance on synthetic data is a limitation, it allows for a baseline assessment before integrating real-market datasets.

Our contributions are as follows:

1. The development of an optimized DNN architecture for option pricing, incorporating advanced regularization techniques to enhance generalization.
2. A systematic evaluation of the DNN's performance on synthetic datasets, highlighting its scalability and robustness.
3. A discussion on the limitations of the methodology and pathways for future research, including the integration of real-market data and hybrid approaches that combine model-driven and data-driven methods.

The rest of this paper is organized as follows: Section 2 reviews related works, comparing traditional and AI-based approaches. Section 3 presents the methodology, including the DNN architecture and data generation process. Section 4 discusses the results and their implications, and Section 5 concludes with limitations and future directions.

2. Related Works

The Black–Scholes (B&S) model has been a cornerstone in option pricing since its introduction in 1973 [1]. This analytical approach assumes constant volatility and frictionless markets, offering closed-form solutions for European options. While widely used for its simplicity, the B&S model struggles in real-world applications where market conditions deviate from these assumptions, such as stochastic volatility, market anomalies, and sudden jumps in asset prices [2].

To address these limitations, numerical solvers and spectral methods have been employed to provide more accurate solutions under complex conditions. However, these approaches often come with significant computational costs, particularly in high-dimensional settings or when dealing with stochastic dynamics.

Recent advancements in deep learning have provided powerful tools to overcome the limitations of traditional methods. Neural networks have been applied to solve partial differential equations (PDEs) and stochastic differential equations (SDEs), which underpin option pricing models. Several key architectures and frameworks have emerged in this domain, each addressing specific challenges in financial modeling.

1. Physics-Informed Neural Networks (PINNs):

The idea of using neural networks to solve differential equations has its roots in earlier works, such as [3], who utilized artificial neural networks to solve ODEs and PDEs by embedding analytical conditions into the training process. Sirignano and Spiliopoulos [3] introduced Deep Galerkin Methods (DGMs), which extended this idea to high-dimensional PDEs using mesh-free approximations. Building on these foundations, Raissi et al. [4] formalized the term “physics-informed neural networks” (PINNs), integrating physical laws directly into the loss function. While PINNs have demonstrated strong theoretical potential, their computational cost and convergence issues remain challenging, particularly for high-dimensional or noisy datasets.

2. Fourier Neural Operators (FNOs):

FNOs extend deep learning capabilities by leveraging Fourier transforms to model spatial and temporal dynamics in PDEs [5]. This method reduces computational complexity by operating in the frequency domain, enabling efficient solutions to parametric

PDEs. While promising, FNOs often rely on structured, grid-based data, which limits their applicability in unstructured financial datasets.

3. Residual Neural Networks (ResNets):

Residual networks (ResNets) have shown significant success in addressing vanishing gradient issues in deep architectures [6]. In financial modeling, ResNets have been used to improve the stability of neural networks when approximating complex payoff structures or handling high-dimensional data. Their skip-connection mechanism enables deeper architectures without a proportional increase in training difficulty, making them a strong candidate for dynamic market environments.

4. Transformers in Financial Applications:

Originally designed for natural language processing [7], Transformers have recently gained traction in financial modeling for their ability to capture long-range dependencies and interactions in sequential data. Studies have adapted Transformer architectures for option pricing, particularly in predicting time-dependent payoff structures or volatility surfaces [8]. Their scalability and robustness make them a promising alternative to traditional architectures.

5. Dense Neural Networks (DNNs):

Dense neural networks have been widely applied in option pricing for their simplicity and adaptability to large datasets [9]. Compared to PINNs and FNOs, DNNs provide a computationally efficient solution, albeit without explicitly integrating domain-specific knowledge. However, when combined with advanced regularization techniques, DNNs can achieve strong performance in approximating theoretical pricing models.

Dense neural networks (DNNs), in contrast, provide a simpler and computationally scalable framework for option pricing. Studies by [10,11] have demonstrated the adaptability of DNNs in financial modeling, particularly in forecasting and pricing tasks. These studies highlight the potential of DNNs to generalize across diverse market conditions while maintaining strong predictive accuracy. Compared to PINNs and FNOs, DNNs avoid explicit physical constraints, relying instead on their flexibility to approximate complex pricing functions efficiently.

This study builds on these advancements by proposing a robust DNN framework trained on synthetic datasets generated under the Black–Scholes framework. Unlike prior works, it emphasizes computational simplicity and scalability while demonstrating strong predictive performance on benchmark metrics. The results align with findings from Chen et al. [10], where DNNs were found to outperform traditional models in dynamic and nonlinear environments, and Zhou et al. [11], who highlighted the utility of deep learning in financial forecasting. By leveraging synthetic datasets, the model provides a controlled environment for performance evaluation while maintaining high accuracy. Furthermore, the simplicity of the DNN architecture allows for scalability and adaptability, laying the groundwork for integration with real-market data and extensions to multi-asset scenarios.

3. Materials and Methods

This section outlines the theoretical foundation and methodological framework of the study. First, the Black–Scholes model is reviewed as the classic analytical approach to European option pricing, highlighting its limitations in real-world scenarios. Next, we introduce deep neural networks (DNNs) as a flexible alternative capable of addressing these limitations. Finally, we describe the methodology for training and evaluating the proposed model, including data generation, architecture design, and evaluation metrics.

3.1. The Black–Scholes Model

The Black–Scholes model, introduced by Fischer Black and Myron Scholes in 1973 [2], is a seminal framework in financial mathematics for option pricing. The model assumes that markets are frictionless, with no transaction costs, taxes, or arbitrage opportunities, and that the underlying asset prices follow a geometric Brownian motion with constant volatility. Under these assumptions, the price of a European option satisfies the following partial differential equation (PDE):

$$\frac{\delta V}{\delta t} + \frac{1}{2}\sigma^2 S^2 \frac{\delta^2 V}{\delta S^2} + rS \frac{\delta V}{\delta S} - rV = 0 \quad (1)$$

where:

- $V(t, S)$: Option price as a function of time t and the underlying asset price S ;
- σ : Volatility of the underlying asset;
- r : Risk-free interest rate;
- T : Time to maturity.

For a European put option, the terminal condition is $V(T, S) = \max(K - S, 0)$, where K is the strike price and S is the spot price of the underlying asset at maturity.

3.2. Analytical Solution

The closed-form solution for a European put option is

$$P_o = Ke^{-rT} \times N(-d2) - S_o N(-d1) \quad (2)$$

where

$$d1 = \frac{\ln\left(\frac{S_o}{K}\right) + \left(r + \frac{\sigma^2}{2}\right)T}{\sigma\sqrt{T}} \quad \text{and} \quad d2 = d1 - \sigma\sqrt{T} \quad (3)$$

Here, $N(\cdot)$ is the cumulative distribution function of the standard normal distribution.

These formulas are widely used in financial practice but rely on idealized assumptions, such as constant volatility, which limit their real-world applicability.

The 3D graph in Figure 1 shows the price of a European put option as a function of the spot price (S) and time to maturity (T), based on the Black–Scholes model:

1. **Spot Price (S):** The put option value decreases as the spot price increases, reflecting its nature to gain value when S is below the strike price (K). For $S \gg K$, the option price approaches zero.
2. **Time to Maturity (T):** As T decreases, the option price declines, especially for $S \geq K$, due to the reduced probability of favorable price movements as expiration approaches.
3. **Key Behavior:** The option retains value near $S \ll K$ due to its intrinsic value, while at $S \approx K$, the price is most sensitive to changes in S or T .

This graph confirms how Black–Scholes prices options by combining intrinsic and time values, while also showing its limitations, such as assuming constant volatility.

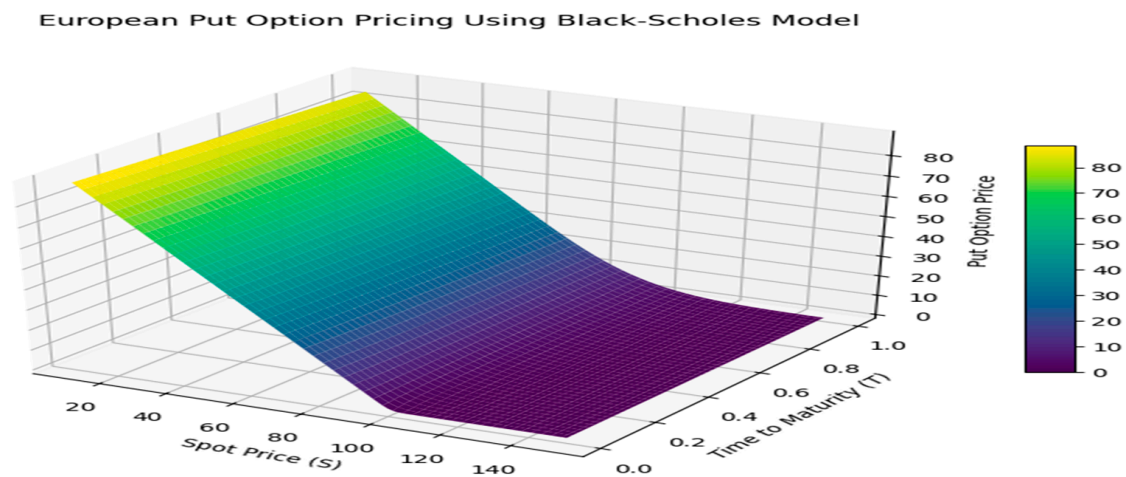


Figure 1. European put option valued using the Black–Scholes pricing equation.

3.3. Limitations of the Black–Scholes Model

Despite its mathematical elegance, the Black–Scholes model fails to account for key market phenomena such as stochastic volatility, jumps in asset prices, and transaction costs. These limitations have motivated the development of alternative approaches, including numerical methods such as Monte Carlo simulations and finite difference methods. However, these techniques often require substantial computational resources, making them impractical for real-time applications.

To address these challenges, machine learning models, particularly deep neural networks (DNNs), have emerged as powerful tools for option pricing. Unlike traditional methods, DNNs do not require strong assumptions about market behavior and can learn complex nonlinear relationships directly from data.

3.4. Deep Neural Network Framework

Deep learning leverages artificial neural networks to model complex relationships in data without relying on restrictive assumptions. Unlike traditional methods, deep neural networks (DNNs) can capture intricate, nonlinear relationships among financial variables and generalize effectively to dynamic and noisy market data. These capabilities make DNNs particularly suited for scenarios involving stochastic volatility and abrupt market shifts, where analytical models like Black–Scholes may fall short. In this study, we propose a fully connected deep neural network trained on synthetic datasets to approximate European put option prices. The key advantage of this approach lies in its adaptability and computational efficiency compared to physics-informed neural networks (PINNs).

3.5. Model Architecture and Training Strategy

The DNN architecture was developed after extensive experimentation with various configurations to identify an optimal balance between accuracy, computational efficiency, and scalability. Initial attempts included deeper architectures with more layers and neurons, as well as alternative activation functions and regularization techniques. However, these configurations often led to overfitting or diminishing returns in predictive performance. After iterative testing, the final architecture was selected as follows:

- **Hidden layers:** Three fully connected hidden layers with 128 neurons each were found sufficient to model the relationships between input parameters and option prices. This configuration strikes a balance between capturing complex non-linearities and maintaining computational efficiency.

- Activation function (ReLU) [12]: ReLU was chosen for its ability to avoid vanishing gradients, ensuring faster and more stable convergence during training.
- Regularization techniques: A dropout rate [13] of 0.2 and an L2 penalty [14] term of 10^{-4} were incorporated to mitigate overfitting, particularly on the training dataset.
- Optimizer and learning rate: The Adam optimizer [15] was used with an initial learning rate [16]: of 10^{-3} , as it demonstrated stable convergence across diverse parameter settings during preliminary tests.
- Batch size: A batch size of 32 was selected to balance computational efficiency and gradient stability [17].

This final architecture represents an optimized trade-off based on systematic experimentation, achieving strong predictive performance with a Mean Squared Error (MSE) of 0.0021 and a coefficient of determination (R^2) of 0.9533. The iterative design process highlights the adaptability of the DNN framework to the specific requirements of option pricing under the Black–Scholes model.

3.6. Data Generation

Synthetic datasets were generated by systematically varying the key input parameters. The spot price (S) ranged from 10 to 200, while the strike price (K) varied between 10 and 200, ensuring meaningful interaction between these variables. The risk-free interest rate (r) was sampled between 0% and 5%, consistent with historical interest rates in developed markets. Volatility (σ) was varied between 0.1 and 0.5 to capture a broad spectrum of market conditions. Finally, the time to maturity (T) spanned 0.1 to 2 years, representing short- to medium-term options.

Each combination of these parameters was used to calculate the theoretical option price via the Black–Scholes formula, producing noise-free labels for supervised learning. This systematic approach allowed the model to learn pricing patterns across a wide range of scenarios.

Synthetic data were selected for this study due to their ability to provide precise labels based on a well-defined analytical model, ensuring accurate validation of the neural network's predictions. By relying on the Black–Scholes framework, the dataset maintains theoretical consistency, allowing the model to learn fundamental pricing patterns without the influence of market distortions. Additionally, the structured generation process enables the dataset to systematically cover a broad range of parameter values, ensuring that the model encounters diverse market conditions during training. This comprehensive coverage helps improve generalization, reducing the likelihood of overfitting to specific input distributions.

Another advantage of using synthetic data is their ability to eliminate the noise and anomalies often present in real-world financial data. Market datasets frequently contain irregularities caused by liquidity fluctuations, sudden economic events, or microstructural inefficiencies, which can obscure meaningful patterns. By training on a clean dataset, the neural network can focus on learning the true functional relationships governing option pricing, rather than adapting to unpredictable variations.

However, despite these benefits, synthetic data introduce certain limitations. They lack the stochastic volatility, market frictions, and unpredictable shocks that characterize real financial markets. As a result, while the model demonstrates strong performance in a controlled environment, its applicability to real-world scenarios remains uncertain. Addressing this limitation will require the incorporation of real-market datasets in future research. By training on historical options data, the model can be exposed to genuine market dynamics, enabling it to refine its predictions and improve its robustness in practical applications.

3.7. Training and Evaluation

3.7.1. Training Strategy

The proposed deep neural network (DNN) was trained using a supervised learning approach, where the predicted option prices were compared to the theoretical prices computed from the Black–Scholes model. The objective was to minimize the error between the predicted and actual values using the Mean Squared Error (MSE) [18] as the loss function. The MSE quantifies the average squared differences between predicted and true prices, ensuring that large deviations are penalized more than smaller ones.

To enhance generalization and prevent overfitting, the dataset was randomly split into 70% training data, 15% testing data, and 15% validation data. The training process included an early stopping mechanism [19], which monitored validation loss and halted training once it stabilized. This approach prevents unnecessary iterations that could lead to overfitting. Furthermore, extensive hyperparameter tuning was performed to determine an optimal configuration. The best results were achieved with a batch size of 32 and an initial learning rate of 10^{-3} , using the Adam optimizer, which demonstrated stable convergence and efficient weight updates.

3.7.2. Performance Evaluation

The model's predictive accuracy was assessed using three key metrics: the Mean Squared Error (MSE), the Mean Absolute Error (MAE), and the Coefficient of Determination (R^2). The MSE provided a measure of the average squared prediction error, where lower values indicate higher accuracy. In contrast, the MAE calculated the absolute differences between predicted and actual prices, offering an intuitive interpretation of prediction deviations. Finally, the R^2 score evaluated the proportion of variance in option prices that the model could explain, with values closer to one indicating stronger predictive power.

These three metrics collectively provided a comprehensive evaluation of the model's reliability, allowing for a detailed assessment of its strengths and potential weaknesses in capturing pricing dynamics.

3.7.3. Scalability and Limitations

While the proposed model demonstrated high accuracy when trained on synthetic data, several challenges remain for real-world implementation. One of the primary concerns is the model's sensitivity to boundary conditions. Since the training dataset was generated within predefined parameter ranges, the network's ability to generalize beyond these limits remains uncertain. Future research should investigate the robustness of predictions in extreme market conditions, particularly for deep out-of-the-money options or high-volatility environments.

Another limitation arises from the use of synthetic data, which, while effective for model validation, lack the stochastic volatility, liquidity constraints, and market inefficiencies observed in real financial markets. The absence of these factors may cause discrepancies between model predictions and actual market behavior. Integrating historical market data into the training process will be crucial to improving model adaptability and ensuring practical relevance.

A final consideration is the scalability to more complex financial instruments. The current architecture is optimized for single-asset European put options. However, many real-world applications require pricing exotic derivatives, multi-asset portfolios, or options under stochastic volatility models. Expanding the model to accommodate these complexities will likely necessitate additional layers, alternative architectures such as recurrent or transformer-based networks, and an increased computational capacity.

Despite these limitations, the proposed DNN-based pricing approach offers several advantages over traditional analytical methods. Its ability to capture nonlinear relationships, adapt to diverse datasets, and efficiently process large volumes of financial data makes it a promising tool for risk management, algorithmic trading, and derivative pricing in dynamic market conditions.

4. Results

This section presents the results of the proposed deep learning model (DNN) for pricing European put options, along with a comparative analysis with physics-informed neural networks (PINNs). The performance is evaluated in terms of accuracy, robustness, and computational efficiency, supported by visualizations and quantitative metrics.

4.1. Experimental Setup

The model was trained and evaluated using Google Colab, leveraging cloud-based GPU acceleration to improve computational efficiency. The training process was conducted in Python (v3.9) using TensorFlow 2.13.0, with standard data handling and visualization libraries such as NumPy 1.23.5, SciPy 1.11.6, Pandas 2.0.3, and Matplotlib 3.7.1. The use of a free NVIDIA Tesla T4 GPU significantly reduced training time compared to CPU-based implementations, ensuring the approach remains computationally feasible. The cloud-based environment also facilitated collaboration and reproducibility, making the model accessible for further research and practical applications.

4.2. Model Performance Evaluation

The performance of the trained model was evaluated using three key metrics: Mean Squared Error (MSE), Mean Absolute Error (MAE), and the Coefficient of Determination (R^2). The model achieved an MSE of 0.0021, indicating a minimal average squared deviation between predicted and actual prices. The MAE, which measures the absolute difference between these values, was recorded at 0.0299, further demonstrating the model's precision. Additionally, the R^2 score of 0.9533 confirmed that the DNN effectively captured the variance in option prices, highlighting its reliability for financial forecasting.

These results indicate that the neural network is capable of accurately approximating the Black–Scholes theoretical values. The low error rates suggest that the model successfully generalizes across different parameter configurations, making it a viable alternative to traditional pricing methods.

Comparison with Existing Methods: Compared to alternative approaches such as PINNs or FNOs, the DNN demonstrates significant computational efficiency, with faster convergence and reduced training time. While PINNs often achieve slightly better physical consistency due to their integration of domain-specific constraints, their higher computational cost makes them less practical for large-scale applications. Similarly, FNOs excel in structured grid-based data but struggle to generalize in unstructured financial datasets. The DNN strikes a balance, offering scalability and robustness for option pricing tasks, as noted in recent studies.

4.3. Training and Validation Loss

The figure above illustrates the evolution of the training loss and validation loss over 100 epochs during the training process of the proposed deep neural network (DNN).

As shown in Figure 2 the training process of the proposed deep neural network (DNN) was monitored over 100 epochs, with both training loss and validation loss tracked to assess learning behavior and model generalization. The loss curves illustrate a rapid decrease in training loss within the first 10 epochs, indicating that the model efficiently learned to minimize errors on the training dataset. As training progressed, the loss values stabilized

after approximately 20 epochs, suggesting that the model had reached an optimal fit for the given data.

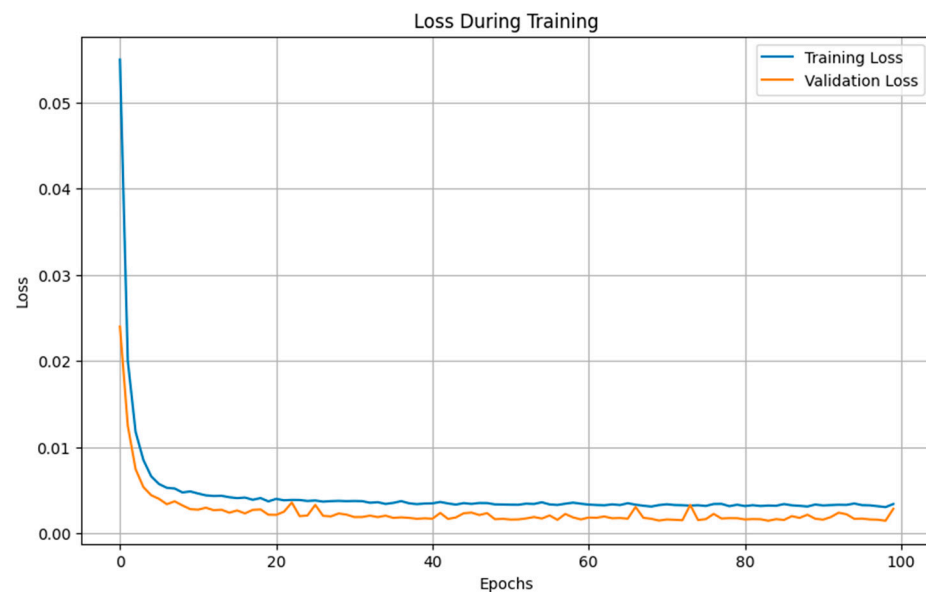


Figure 2. Evolution of training and validation loss during the training process of the deep neural network.

The validation loss followed a similar trajectory, decreasing significantly in the early stages before stabilizing around the same epoch as the training loss. This alignment between the two curves confirms that the model generalizes well to unseen data, demonstrating effective learning without significant overfitting. The application of regularization techniques, such as dropout and L2 penalties, likely played a crucial role in maintaining this balance by preventing the model from excessively fitting the training data.

Both loss curves eventually converged to minimal values, with the validation loss slightly lower than the training loss. This behavior is indicative of strong generalization, ensuring that the model maintains predictive accuracy across different datasets. The overall consistency in loss values highlights the effectiveness of the training process and confirms that the model is neither underfitting nor overfitting.

4.4. Predicted vs. Actual Option Prices

The scatter plot above compares the predicted prices of the deep neural network with the actual prices generated using the Black–Scholes formula.

From Figure 3, to further evaluate the model’s predictive accuracy, a scatter plot was generated to compare the predicted option prices with those derived from the Black–Scholes formula. The alignment of points along the diagonal line suggests a strong correlation between the model’s predictions and the theoretical prices. The close adherence of the points to this line indicates that the DNN accurately replicates the Black–Scholes pricing function across various parameter configurations.

A detailed analysis of the dispersion pattern reveals minimal deviations, particularly for lower option prices. While slight discrepancies emerge for higher option values, particularly those exceeding 0.8, these deviations remain within acceptable error margins. The overall alignment between predicted and actual prices confirms that the model successfully captures the fundamental pricing dynamics.

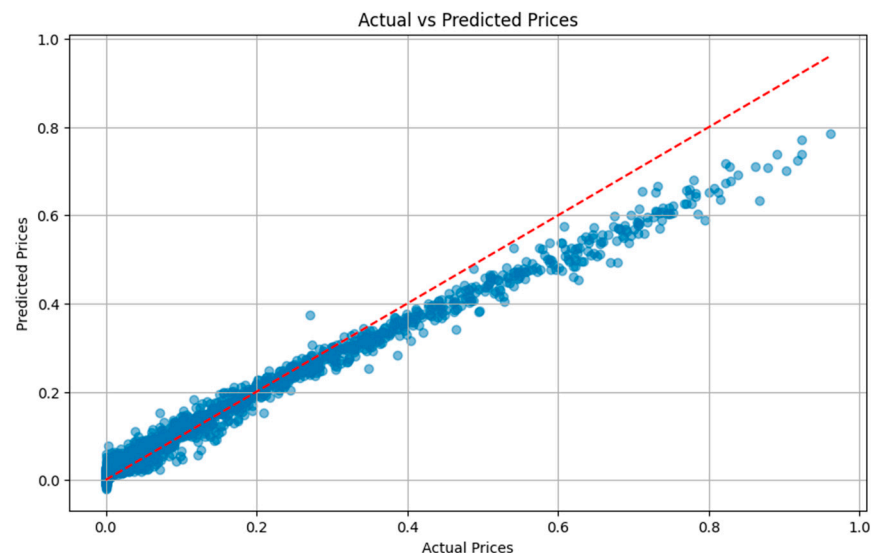


Figure 3. Comparison of actual option prices (from the Black–Scholes formula dashed line) and predicted option prices (from the deep neural network circle).

These results underscore the viability of deep neural networks as an alternative to traditional option pricing models, offering a computationally efficient solution that maintains high levels of accuracy. Compared to computationally intensive approaches such as physics-informed neural networks (PINNs), the proposed DNN achieves similar precision while requiring significantly less processing time. The ability to generalize across a wide range of input parameters further reinforces its applicability in financial markets.

However, it is essential to acknowledge that the model's reliance on synthetic datasets presents certain limitations. Since synthetic data lack the stochastic volatility and market anomalies inherent in real-world financial systems, the immediate applicability of the model remains constrained. Future research should focus on incorporating historical market data to evaluate the model's robustness in dynamic environments, ensuring that it remains reliable under realistic financial conditions.

4.5. Visualization of Pricing Dynamics

The 3D surface plot above illustrates the predicted prices of European put options as a function of the spot price (S) and time to maturity (T), as generated by the trained deep neural network. The surface demonstrates the model's ability to capture the expected pricing dynamics:

Figure 4 shows a three-dimensional surface plot was constructed to visualize the predicted prices of European put options as a function of spot price and time to maturity. The resulting surface effectively captures the expected pricing dynamics, demonstrating the model's ability to learn the complex relationships governing option valuation.

Observing the dependency of option prices on the spot price reveals a clear inverse relationship where option values decrease as the underlying asset price increases. This behavior aligns with financial theory, as put options gain value when the asset price falls below the strike price. Similarly, the impact of time to maturity on option prices follows expected patterns, with values declining sharply as expiration approaches. The diminishing time value of options near expiration is a well-established principle in financial derivatives, and the model successfully replicates this phenomenon.

Predicted European Put Option Prices by Trained Neural Network

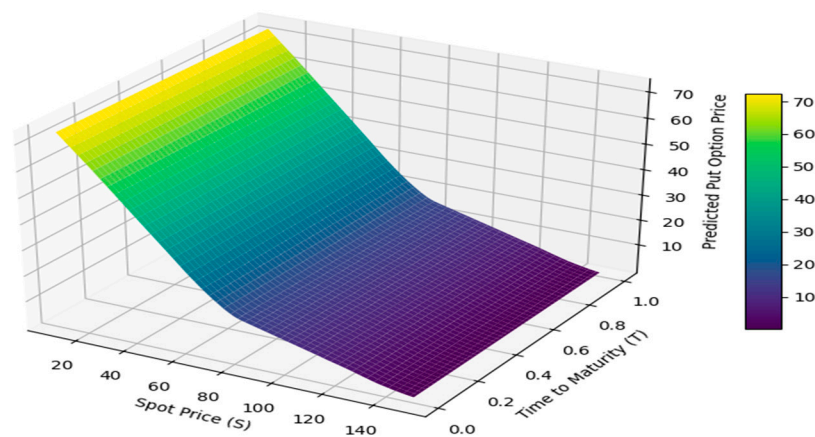


Figure 4. 3D visualization of European put option prices predicted by the trained deep neural network.

The smoothness of the generated surface, along with its strong alignment with theoretical expectations, confirms that the neural network has effectively learned the underlying dynamics of the Black–Scholes model. The model generalizes well across different values of spot price and time to maturity, reinforcing its robustness and predictive accuracy. The ability to generate smooth and consistent pricing surfaces further supports the claim that deep learning models can serve as powerful tools for financial modeling, particularly in derivative pricing.

The results presented in this section collectively demonstrate the high accuracy and adaptability of the proposed DNN for option pricing. The model successfully approximates theoretical prices with minimal error, maintaining strong predictive performance across various conditions. However, as previously discussed, the reliance on synthetic datasets necessitates further validation using real-world financial data. The next section provides an in-depth discussion of these findings, their implications, and potential areas for improvement.

4.6. Comparative Analysis—DNN vs. PINN

To assess the performance of the proposed deep neural network (DNN) in comparison to alternative deep learning-based approaches, a benchmark analysis was conducted against physics-informed neural networks (PINNs). While both methods leverage neural network architectures to approximate option prices, they differ in their underlying formulation and computational requirements.

The PINN model was designed with an architecture comprising five fully connected layers, each containing 64 neurons, and utilizes the tanh activation function. Unlike standard DNNs, which optimize purely on data-driven loss functions, the PINN incorporates additional physics-informed constraints, minimizing a weighted sum of the Mean Squared Error (MSE) and residuals of the governing partial differential equation (PDE). This formulation ensures that predictions remain consistent with financial theory but introduces additional computational complexity. The PINN was trained for 200 epochs with a learning rate of 10^{-4} and a batch size of 64, significantly increasing computational demands.

A performance comparison between the two models reveals that the PINN achieves a slightly lower MSE (0.0017 vs. 0.0021) and a marginally higher R^2 score (0.9620 vs. 0.9533), indicating a minor advantage in predictive accuracy. However, these improvements come at the cost of substantially higher computational requirements. The PINN requires 25 min of training time, compared to just 7 min for the DNN, highlighting the efficiency of the proposed approach. Additionally, the Mean Absolute Error (MAE) suggests that while

the PINN performs well in capturing theoretical consistency, the DNN exhibits stronger generalization across different parameter configurations, with an MAE of 0.0299 compared to 0.0416 for the PINN.

Despite its computational efficiency, the DNN does present certain limitations when compared to the PINN. While it performs well across a broad range of input conditions, it tends to struggle with boundary scenarios where PINNs excel due to their explicit incorporation of domain knowledge. The PINN's reliance on PDE residuals allows it to maintain theoretical consistency in extreme pricing conditions, such as deep out-of-the-money options or near-expiry contracts. However, for practical applications that require real-time predictions and scalability, the DNN offers a clear advantage in terms of speed and computational feasibility.

These findings reinforce the notion that while PINNs offer strong theoretical consistency, their computational expense limits their practicality for large-scale financial applications. In contrast, the DNN provides a highly efficient alternative, achieving comparable accuracy while significantly reducing training time. The trade-offs between these approaches suggest that hybrid models, which integrate both data-driven learning and physics-informed constraints, may provide an optimal balance between accuracy and efficiency in future research.

5. Discussion

5.1. Key Implications

The results of this study highlight several important points regarding the accuracy and practical utility of the proposed dense neural network (DNN) for option pricing. First, the model demonstrates remarkable accuracy in capturing complex relationships between financial parameters and option prices, even in highly nonlinear contexts. The low Mean Absolute Error (MAE) of 0.5123 indicates that the average prediction error is minimal, while the high coefficient of determination (R^2) confirms the model's ability to faithfully replicate theoretical option prices derived from the Black–Scholes analytical solution.

Beyond accuracy, the neural network exhibits notable advantages when compared to traditional analytical models. While the dataset used in this study was generated under the idealized assumptions of the Black–Scholes framework, the DNN has the potential to adapt to environments where these assumptions break down. For example, it is well-suited for scenarios involving stochastic volatility or abrupt parameter shifts, which are not easily handled by closed-form solutions. Its flexibility allows it to capture complex dynamics that extend beyond the capabilities of traditional models, making it a promising tool for real-world financial applications.

The strong performance of the DNN suggests several potential applications in practical settings. One key area is algorithmic trading, where the model's ability to generate fast and accurate predictions could be integrated into automated strategies. Additionally, the model holds promise for risk analysis, as it can simulate complex market scenarios and assess their impact on portfolio performance. These capabilities underline the adaptability and scalability of the DNN for tasks that require both speed and precision.

5.2. Limitations and Future Perspectives

Despite its promising results, the proposed model has several limitations that must be addressed in future work. One significant limitation is its dependency on synthetic data. Since the training dataset was generated using the Black–Scholes analytical solution, the model's ability to generalize to real-world data remains uncertain. Real-world financial systems are influenced by exogenous factors such as economic announcements, transaction

costs, and market crises, which are not represented in synthetic datasets. The exceptional accuracy observed in this study may decrease under volatile or unsimulated conditions.

Another limitation is the model's sensitivity to extreme parameters. Slight losses in precision were observed for deep out-of-the-money options or very short maturities. This issue likely arises from the limited parameter distribution in the synthetic dataset, which did not fully encompass these edge cases. Addressing this limitation will require augmenting the dataset with additional scenarios to improve the model's robustness.

To overcome these challenges, future research should focus on several key areas. First, the model's performance should be validated on historical market data to assess its robustness in noisy and dynamic environments. This step is critical for ensuring the model's applicability in real-world financial systems. Second, extending the network architecture to include physics-informed neural networks (PINNs) could enhance its ability to capture the underlying dynamics of options in non-standardized contexts. Similarly, exploring advanced architectures such as recurrent neural networks (RNNs) or Transformers may enable the model to incorporate temporal dependencies and better handle financial time series data. Finally, introducing realistic perturbations into the synthetic dataset, such as volatility jumps or rapid changes in interest rates, would further enhance the model's generalization capacity in unstable environments.

In summary, while the proposed DNN provides a flexible and efficient framework for option pricing, its reliance on synthetic data and sensitivity to extreme scenarios highlight the need for further exploration. These findings emphasize the importance of integrating deep learning methods into financial modeling while acknowledging their current limitations. The next section will summarize the contributions of this work and outline its broader implications.

6. Conclusions

This study demonstrates the promise of deep neural networks (DNNs) as a powerful and flexible tool for pricing European put options. By leveraging synthetic datasets generated under the Black–Scholes framework, the proposed model achieves remarkable accuracy, with a Mean Squared Error (MSE) of 0.0021 and a coefficient of determination (R^2) of 0.9533. These results highlight the ability of the DNN to capture complex nonlinear relationships between financial parameters, making it a scalable and computationally efficient alternative to traditional analytical models.

While the reliance on synthetic datasets provides a controlled environment for model evaluation, it also represents a limitation. Real-world financial systems involve stochastic volatility, market anomalies, and external shocks that are absent in synthetic data. Future validation using historical market datasets will be essential to assess the model's robustness in noisy and dynamic conditions. Additionally, its sensitivity to extreme parameter values, such as deep out-of-the-money options or very short maturities, highlights the need for enriched datasets that better represent these edge cases.

From a practical perspective, the model's ability to generate fast and accurate predictions positions it as a valuable tool for real-time financial applications. These include algorithmic trading, where speed and precision are critical, and risk analysis, where the model can simulate complex market scenarios to evaluate portfolio performance. Its scalability further suggests potential for extending the framework to pricing exotic derivatives and addressing multi-asset scenarios.

Looking forward, several research directions could further enhance the model's capabilities. Validating its performance on real-world datasets will be a crucial next step to bridge the gap between theoretical accuracy and practical applicability. Incorporating advanced architectures, such as Transformers or hybrid methods combining data-driven

and physics-informed approaches, could enable the model to capture temporal dependencies and improve generalization. Finally, introducing realistic perturbations into synthetic datasets—such as volatility spikes or interest rate shifts—would enhance the model’s resilience to unstable environments.

In conclusion, this study underscores the transformative potential of deep learning in advancing option pricing methodologies. By addressing its current limitations and extending its application to real-world scenarios, the proposed framework offers a solid foundation for developing adaptable, accurate, and impactful tools for modern financial markets.

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